THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 8: **Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The time between two consecutive compressions at a point is the: 1

(a) speed **(b)** wavelength **(c)** time period **(d)** amplitude

2. A sound of 20 dB compared with a sound of 40 dB is: 1

(a) less loud **(b)** inaudible **(c)** more loud **(d)** equally loud

3. The number of waves produced in 5 s by a source of 20 Hz is: 1

(a) 20 **(b)** 25 **(c)** 4 **(d)** 100

4. A shrill sound has: 1

(a) low pitch **(b)** high pitch **(c)** large wavelength **(d)** low frequency

5. In 10 s a source produces 4000 vibrations. The frequency is: 1

(a) 400 Hz **(b)** 4010 Hz **(c)** 0.0025 Hz **(d)** 40000 Hz

6. The threshold of hearing is about: 1

(a) 120 dB **(b)** 60 dB **(c)** 200 dB **(d)** 0 dB

7. A boy hears an echo 1.5 s after shouting (v = 340 m/s). The distance of the cliff is: 1

(a) 340 m **(b)** 170 m **(c)** 255 m **(d)** 510 m

8. A man's voice compared with a woman's voice has: 1

(a) the same frequency **(b)** lower frequency
(c) higher frequency **(d)** higher pitch

9. A source with a frequency of 10 Hz produces: 1

(a) an ultrasonic wave **(b)** an audible wave
(c) an infrasonic wave **(d)** a light wave

10. Sound levels above 120 dB can cause: 1

(a) no effect **(b)** better hearing
(c) pleasant feeling **(d)** pain and hearing damage

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. State why the sound of a flute is different from that of a violin even at the same pitch. 2

12. Explain why we can talk through a string telephone. 2

13. Find the minimum distance of a reflecting surface for a distinct echo at 20 degrees Celsius (v = 344 m/s). 2

14. Find the frequency of sound of wavelength 0.34 m in air (v = 340 m/s). 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A boy hears the echo of his shout after 2 s while standing in front of a cliff. If the speed of sound increases to 350 m/s, the new time is? Take the distance as 340 m. 3

16. Explain why the speed of sound in a solid is greater than in a gas using the elasticity and density. 3

17. Why do we hear a sound inside a closed room louder than in the open? 3

OR

A sound wave of frequency 600 Hz travels at 360 m/s. Find the wavelength and the distance between a compression and the next rarefaction. 3

18. Why does sound travel better in the night than in the day? Give a reason. 3

OR

Explain the use of sound waves in finding flaws in metal castings. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A sonar on a ship sends a signal which returns after 0.8 s from a submarine. Find the distance of the submarine. If the signal's frequency is 40 kHz, find the wavelength (v = 1500 m/s) and the number of waves in the pulse of 0.01 s. 5

OR

Explain how the depth of a sea can be found by SONAR and how a fish-finder works. A fish-finder gets an echo from a fish after 0.04 s in water (v = 1500 m/s); find the depth of the fish. 5

20. Describe how to find the speed of sound in air using an echo method with a stopwatch and measuring tape. Include the calculation for d = 85 m and time for 10 echoes. 5

OR

Explain why sound is not heard in a vacuum with the bell jar experiment, and discuss how astronauts communicate in space. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND